# Course project detail

**Project title:** ML-Powered-Medical-Diagnosis-Assistant

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**Problem definition:** In rural or resource-limited areas, timely medical diagnosis is hampered by a shortage of qualified professionals and limited infrastructure. Reliance on vague symptom descriptions causes misdiagnosis or delayed treatment. Existing mobile health solutions lack offline functionality, voice interaction, and robust real-time clinical data integration.
This project aims to use **machine learning models** to generate accurate, data-driven diagnosis predictions and validate these using real-world clinical datasets.

**Solutions expected:**

- Use **machine learning models** trained on symptom-disease datasets for accurate diagnosis prediction with confidence scores.
- Implement **Apache Spark** for real-time processing and validation of Electronic Health Records (EHR) data to support ML predictions.
- Develop a **native Android app** that accepts text and offline voice inputs for symptoms, enabling offline usage.
- Provide a clear diagnosis report including symptoms matched, confidence score, similar EHR cases, suggested actions, and urgency level.
- Ensure local **data logging** for offline analytics and model improvement.
- Include **statistical validation** for diagnosis accuracy and model performance.
- Incorporate **privacy and security** safeguards for sensitive medical data.
- Use visualization for symptom trends and diagnosis accuracy to monitor system performance.

## Technologies detail:

### 1. Machine Learning Models for Diagnosis

- Use supervised ML models trained on disease-symptom datasets (e.g., gradient boosting, random forests, or neural networks) to interpret symptom inputs and predict probable illnesses.
- Employ specialized biomedical pretrained models (MedLM, MedGemma, or BioGPT) to extract symptom features and support diagnosis generation with learned clinical knowledge embeddings.

**2. Real-Time Data Processing with Apache Spark**

- Utilize Spark for real-time ingestion, processing, and filtering of large-scale EHR datasets (such as MIMIC-IV or eICU).
- Perform pattern recognition and validation of ML diagnosis outputs by matching patient-specific demographics and clinical histories.

**3. Android Application Interface**

- Develop a native Android app supporting both text and **offline speech-to-text conversion** for symptom input.
- Implement UX focused on symptom reporting, diagnosis feedback, and local logging for further ML model refinement.

**4. Data Analytics and Statistical Validation**

- Validate ML model predictions statistically using modules like SciPy and scikit-learn for accuracy, confidence scoring, and performance tracking.
- Visualize symptom trends and diagnosis accuracy with libraries such as Matplotlib and Seaborn.

**Block diagram and Flowchart**:

1. **User Input**
   - Text or voice symptoms entered via Android app
2. **Preprocessing**
   - Voice converted to text (if applicable)
   - Text tokenized, cleaned, normalized
3. **ML Diagnosis Module**
   - ML models generate probable diagnoses and confidence scores based on symptom input
4. **EHR Data Validation (Apache Spark)**
   - Real-time filtering and pattern matching of EHR datasets to validate ML predictions
5. **Diagnosis Output Module**
   - Combined diagnosis ranking, confidence, suggestions, urgency level
6. **Android Display & Logging**
   - Display results to user
   - Log data locally for offline analytics and future retraining

**Flowchart**

1. **Start**
2. User opens app and inputs symptoms (text/voice)
3. Convert voice to text (if applicable)
4. Preprocess input: tokenize, clean, normalize
5. Pass symptoms to ML model for diagnosis generation
6. Retrieve and process relevant EHR data with Spark for validation
7. Combine ML results and EHR evidence, rank diagnoses
8. Display diagnosis, confidence, recommended actions, and urgency
9. Log interaction locally for offline use
10. **End**

**Algorithm details:**

1. User inputs symptoms via text or offline voice in the Android app.
2. Preprocessing converts speech to text (if applicable), tokenizes, and normalizes symptom data.
3. The ML model trained on symptom-disease mappings predicts probable diagnoses and respective confidence scores.
4. Apache Spark filters and analyzes relevant EHR patient records to cross-validate ML predictions using demographic and clinical pattern matching.
5. ML outputs and EHR validation scores are merged to yield a final ranked diagnosis list.
6. Results including diagnosis, symptom break-down, confidence, and actionable steps are displayed with an urgency level indicator.
7. User interaction data is logged locally for incremental retraining and model improvement.

**Input datasets details: Datasets and Training Data**

| **Dataset Name** | **Description** | **Access Type** | **ML Use Case** |
|---|---|---|---|
| Disease and Symptoms Dataset | Maps diseases to symptoms in CSV format | Free & Open | Training supervised ML classifiers |
| MIMIC-IV | ICU patient EHR data with vitals and diagnoses | Credentialed Access | Feature extraction, validation |
| AIMI Dataset Index (Stanford) | Multimodal data including imaging and EHR | Mixed Access | Advanced ML feature integration |

| Dataset Name | Description | Access Type | ML Use Case |
|---|---|---|---|
| Open Disease Symptoms (Kaggle) | CSV of symptoms and diseases | Free & Open | Model prototyping |

**Expected output and finding:**

- **Diagnosis:** ML model prediction of primary condition (e.g., "Acute Bronchitis").
- **Confidence Score:** Probability from classifier and Spark EHR validation combined for user trust.
- **Symptom Breakdown:** Explanation of symptoms contributing to the diagnosis.
- **EHR Evidence:** Number of matched clinical cases supporting prediction.
- **Suggested Action:** Clear, ML-based recommendations based on diagnosis severity.
- **Urgency Level:** Severity classification to prioritize clinical attention.
- **Disclaimer:** Legal note about model limitations.

**Conclusion:** This ML-powered Medical Diagnosis Assistant leverages **machine learning algorithms** trained on biomedical and clinical data, integrated with Spark-based large-scale EHR validation, to provide offline-capable, voice-interactive, and data-driven medical diagnosis for resource-constrained environments. It empowers patients with reliable, interpretable diagnostic support and actionable insights without requiring constant internet connectivity or immediate medical professional intervention.

**Google Scholar (2 to 5) references (closely related to the project):**


1. Tiwari, A., Manthena, M., Saha, S., Bhattacharyya, P., Dhar, M. *Dr. Can See: Towards a Multi-modal Disease Diagnosis Virtual Assistant,* Proceedings of the 31st ACM International Conference on Information Retrieval, 2022. Google Scholar Link
2. Tiwari, A., Saha, T., Saha, S., Bhattacharyya, P., Begum, S., Dhar, M. *Symptoms Are Known by Their Companies: Association-Guided Disease Diagnosis Assistant,* BMC Bioinformatics, 2022. Google Scholar Link
3. Yang, D., Wei, J., Li, M., Liu, J., Hu, M., He, J., Zhou. *MedAide: Information Fusion and Anatomy of Medical Intents via LLM-based Agent Collaboration,* WarXiv preprint, 2025. Google Scholar Link

4. Varas, J., Coronel, B.V., Villagrán, I., Escalona, G., Hernandez, R., Schuit, G. *Innovations in Surgical Training: Exploring the Role of Artificial Intelligence and Large Language Models (LLM),* Revista do Colégio Brasileiro de Cirurgiões, 2023.
   Google Scholar Link
5. Tiwari, A., Saha, S., Bhattacharyya, P. *A Knowledge-Infused Context-Driven Dialogue Agent for Disease Diagnosis Using Hierarchical Reinforcement Learning,* Knowledge-Based Systems, 2022.
   Google Scholar Link